24. In $CaCO_3(S) \rightarrow CaO(S) + CO_2(g)$, entropy change is positive. **in option d, A solid reactant gives a gaseous product. Hence the entropy change is expected to be maximum for this process.**

25.

$\Delta G = \Delta H - T \Delta S$

At 300K

$\Delta G = 30000 \text{ J mol}^{-1} - 300 \text{ K}$

$\times 100 \text{ J K}^{-1} \text{ mol}^{-1}$

$\Delta G = 0$

above 300 K ; ΔG will be negative and reaction becomes spontaneous.

I Keys to the short answer questions:

53.

SOLUTION :

Given :

$n = 2$ moles

$V_i = 500 \text{ ml} = 0.5 \text{ lit}$

$V_f = 2 \text{ lit}$

$T = 25^0C = 298 \text{ K}$

$w = -2.303 \text{ nRTlog} \left(\frac{V_f}{V_i} \right)$

$w = -2.303 \times 2 \times 8.314 \times 298$

$\times \log \left(\frac{2}{0.5} \right)$

$w = -2.303 \times 2 \times 8.314$

$\times 298 \times \log(4)$

$w = -2.303 \times 2 \times 8.314$

$\times 298 \times 0.6021$

$w = -6871 \text{ J}$

$w = -6.871 \text{ kJ}$

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